

Abdelwaheb Hafs, PhD

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Robotics researcher specializing in human-robot interaction, assistive control, and predictive/game-theoretic control for exoskeletons. My work develops computational control methods for personalized movement assistance and combines modeling, simulation, experimental validation, and interdisciplinary collaboration between robotics and human movement science.

Research interests: Robot learning, adaptive control, model-based robotics, human-robot interaction, robot manipulation.

Education

Université Paris-Saclay

Paris, France

Ph.D. in Sport and Human Movement Sciences

2021-2025

- **Dissertation:** *Game-theoretic predictive control for intuitive movement assistance in human-robot interaction*
- **Award:** Demeny-Vaucanson Prize, 2022

Université Paris-Saclay

Paris, France

Master's Degree in Robotics, Assistance, and Mobility

2020-2021

- **Master's thesis:** Adaptive Control Laws for Improving Human-Exoskeleton Interaction
- **Rank:** 1st out of 10 students

National Polytechnic School

Oran, Algeria

Bachelor's and Master's Degrees in Automatic Control

2017-2020

- **Master's thesis:** Predictive Control Applied to a Two-Wheeled Mobile Robot
- **Rank:** 2nd out of 29 students

School of Electrical and Energetic Engineering of Oran

Oran, Algeria

Preparatory Classes

2015-2017

- **Rank:** 3rd out of 395 students

Academic Positions

Faculty of Sport Sciences, Université Paris-Saclay

Paris, France

Temporary Teaching and Research Associate (ATER)

Sep. 2025-Aug. 2026

- **Research:** Implementation of intuitive assistive control laws with trajectory prediction for the study of human-robot interaction, in collaboration with CIAMS and INRIA
- **Teaching:** 192 hours in Biomechanics (L1 STAPS), Robotics (M1 ISMH), and Programming with Python (M2 ISMH)

CIAMS, Université Paris-Saclay

Paris, France

Ph.D. Researcher

Oct. 2021-Dec. 2025

- Conducted research on game-theoretic and predictive control for human-robot interaction
- Supervised undergraduate/master's interns involved in research projects
- Taught 192 hours in Biomechanics, Robotics, and Programming

LURPA (ENS Paris-Saclay) and CIAMS (Université Paris-Saclay)

Paris, France

Master's Research Internship

Mar. 2021-Sep. 2021

- Developed a control system based on adaptive oscillators
- Identified the dynamics of a coupled system
- Used Qualisys Track Manager for motion analysis experiments

Research Expertise

Human-robot interaction, Assistive robotics and exoskeletons, Model predictive control, Differential games / game-theoretic control, Adaptive control, Robot modeling and simulation, Motion analysis / experimental validation, Python, MATLAB, C++, C.

Publications

- **Hafs, A.**, Verdel, D., Burdet, E., Bruneau, O., & Berret, B. (2024). Finite-horizon inverse differential game approach for optimal trajectory-tracking assistance with a wrist exoskeleton. *BioRob 2024*. DOI: 10.1109/BioRob60516.2024.10719810
- **Hafs, A.**, Verdel, D., Bruneau, O., & Berret, B. (2025). Game-theoretic interaction control for assistive exoskeletons: a 2-DOF simulation study. *IFAC-PapersOnLine*. DOI: 10.1016/j.ifacol.2025.10.231
- **Hafs, A.**, Farr, A., Verdel, D., Bruneau, O., Burdet, E., & Berret, B. (2026). Model predictive game control for personalized and targeted interactive assistance. *Communications Engineering*. DOI: 10.1038/s44172-026-00605-8
- **Hafs, A.** (2025). Game-theoretic predictive control for intuitive movement assistance in human-robot interaction. Ph.D. dissertation, Université Paris-Saclay.

Conference and Workshop Communications

- **Hafs, A.**, Verdel, D., Jerray, J., Bruneau, O., Vignais, N., Berret, B., & Fribourg, L. (2023). Convergence and robustness of the Hopf oscillator applied to an ABLE exoskeleton: reachability analysis and experimentation. *Modélisation des Systèmes Réactifs (MSR'23)*, Toulouse, France.
- **Hafs, A.**, Verdel, D., Gomes, W., Bruneau, O., & Berret, B. (2023). Optimizing human-robot interactions through differential game control. Workshop on Multilimb Coordination in Human Neuroscience and Robotics: Classical and Learning Perspectives, IROS 2023, Detroit, USA.
- Société Francophone Posture Équilibre et Locomotion (SOFPEL), 2024, Paris, France.
- GDR Day on “Contrôle sensori-moteur pour la robotique”, 2025, Paris, France.
- Biomeca@Inria Day, 2026, Paris, France.

Internship Supervision

- **Spring 2023** Supervision of an internship on the comparative study of Model Predictive Control and differential game-based control for human-exoskeleton interaction.
- **Feb.-Jul. 2024** Supervision of an internship on the analysis of collaborative human-exoskeleton interaction based on differential game theory.
- **Spring 2025** Supervision of an internship on the analysis of the planning horizon in collaborative human-exoskeleton interaction based on differential game theory.

Reviewing Activity

- Reviewed one abstract for IFAC 2025.